

John Chambard

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EDUCATION

Purdue University | *B.S. Computer Science, Concentrations: Software Engineering & Cybersecurity* May 2028
Relevant Coursework: Computer Architecture, Discrete Mathematics, Operating Systems

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript, Golang, Rust, OCaml, SQL, HTML/CSS, MASM Assembly
Frameworks & Libraries: React, React Native, Node.js, Flask, FastAPI, TensorFlow, PyTorch, scikit-learn, Expo, Vite, Firebase, Supabase, REST APIs
Developer Tools: Git, GitHub, Docker, AWS (Lambda, EC2, S3, CloudWatch), Linux, CUDA, Jira, VS Code

EXPERIENCE

Software Engineer Intern | *Phillips 66* May 2026 - August 2026

- Building full-stack invoicing SAP admin tool serving 25,000+ customers handling \$2M+ in transactions per day
- Refactoring tech debt on 4 repos and migrating 3 on-prem tools to cloud infrastructure, improving offsite refinery response times
- Developing internal AI agent that parses refinery documents and gives weekly digests for leadership

Software Engineer Intern | *NETSYNC* May 2025 - January 2026

- Architected EC2-based cloud infrastructure for VM provisioning, cutting cybersecurity lab setup time from days to under 3 hours for 12+ universities, using Lambda automation to enable scalable, distributed on-demand deployment
- Led app development adopted by 87+ clients to manage AWS migrations averaging \$12K in cloud cost reductions
- Worked within a 20-person international Agile Scrum team, shipping to Fortune 500 and government clients

PROJECTS

JackScript - Custom Programming Language | *Java, Compiler Design, Interpreter* August 2024 - May 2025

- Engineered a complete programming language from scratch: lexer, parser, AST, interpreter, and runtime. Supporting 6+ data types and 20+ built-in functions, parsing at 10,000+ tokens/second
- Adopted by school UIL competitive programming team for practice and used in 20+ programs
- Reduced competitive programming solution boilerplate by 10% through careful language design decisions

HearLink - Sound Detection Wearable | *C/C++, Python, ESP32-S3, YAMNet, PCB Design* January 2026

- Engineered real-time embedded pipeline on ESP32-S3 reading 4 microphones over dual I2S buses with 550ms end-to-end latency; achieved 70% classification accuracy across 521 sound classes and direction within 7 degrees
- Implemented delay-and-sum beamforming for multi-microphone sensor fusion and real-time YAMNet inference over WiFi streaming; designed and fabricated custom PCBs with PWM-tuned piezo haptic feedback circuits, delivered in 30 hours at StarckHacks 2026
- Invited by judges to author research article on performing beamforming on ESP32-S3 architecture

The Bear Den | *React Native, TypeScript, Golang, Firebase, REST APIs* May 2023 - May 2025

- Deployed production app reaching 6,000+ users across 10+ schools within 6 months; 4.9-star App Store rating
- Designed Golang REST API for sensitive student data handling 3,000+ requests/day with sub-100ms response
- Implemented A* pathfinding across 4 floors and 100+ nodes, running 30% faster than alternative solutions

Agentic OS - Personal AI Life System | *SQL, pgvector, Redis, Ollama, CUDA, Linux* April 2026 - Present

- Self-hosted local AI system on custom-built dual-boot Linux/Windows PC (Nvidia GPU); configured CUDA drivers, cuDNN, and NVIDIA Linux toolchain from scratch for fully GPU-accelerated LLM inference
- Engineered RAG pipeline over 12,000+ personal documents; built nightly AI scheduler with fitness and networking coach, cutting daily planning time by 90% (16 min → 90 sec), running fully local via Ollama

Tutelum - Agentic Payment Governance | *x402 Protocol, CDP SDK, MCP, Ollama* March 2026 - April 2026

- Created startup MVP enabling an autonomous AI agent to execute and track payments through direct crypto transfer and the x402 protocol from a provisioned on-chain wallet, built on the Coinbase CDP SDK
- Built policy layer governing agent spend: per-transaction and daily limits with whitelist/blacklist support
- Exposed payment and policy actions as tools over an MCP server and drove them with a local Ollama agent

LEADERSHIP

Project Manager | *STACK - Purdue Software Engineering Club* January 2026 - Present

- Led 6-person team across 8 sprints with 40+ PRs merged, delivering a peer-matching platform for a nonprofit client
- Promoted after 1 semester from a pool of 50 developers, driving sprint planning, code reviews, and architecture